

Project Design Phase
Solution Architecture

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Team ID	
Project Name	
Maximum Marks	4 Marks

Solution Architecture

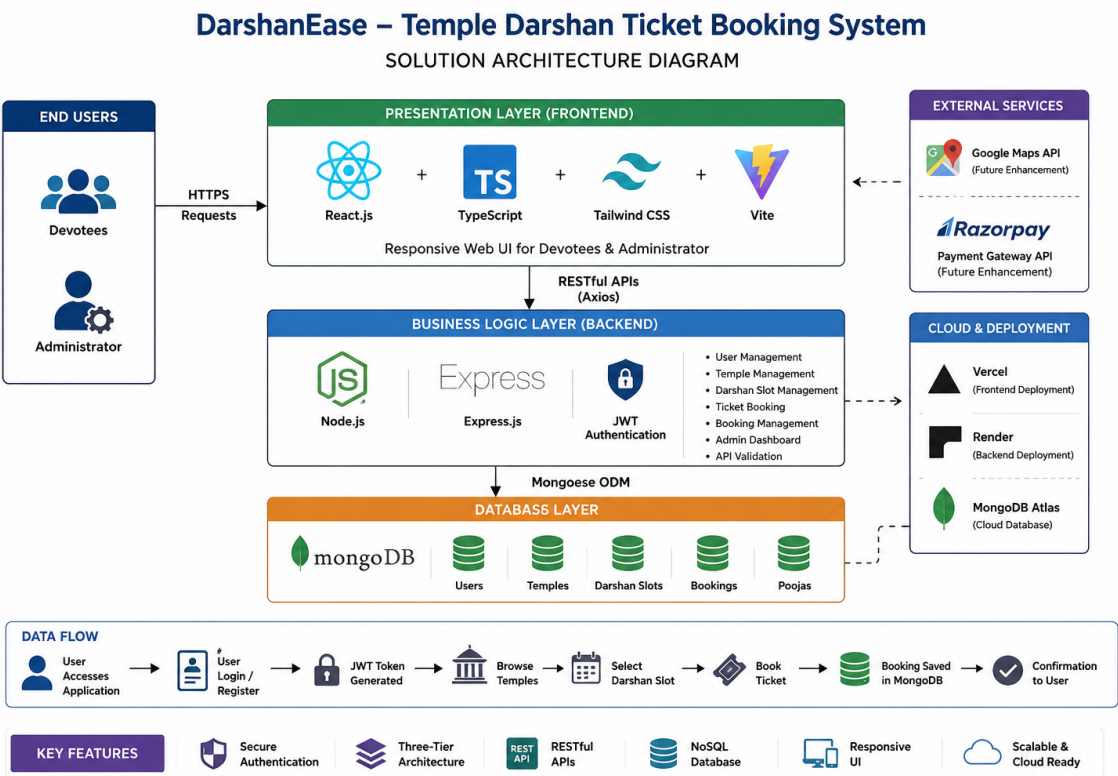
Overview

The **DarshanEase – Temple Darshan Ticket Booking System** follows a **Three-Tier MERN Architecture** that separates the application into the **Presentation Layer**, **Business Logic Layer**, and **Database Layer**. This architecture ensures modularity, maintainability, scalability, and secure communication between the client and server.

The React.js frontend provides an interactive user interface for devotees and administrators. User requests are sent through RESTful APIs to the Express.js backend, where business logic, authentication, booking management, and temple management are processed. The backend communicates with MongoDB to store and retrieve application data.

This architecture enables secure authentication using JWT, efficient database operations using Mongoose, and cloud deployment using modern hosting platforms such as Vercel, Render, and MongoDB Atlas.

Solution Architecture Diagram



Architecture Components

1. Presentation Layer

The presentation layer provides the graphical user interface for both devotees and administrators.

Technologies Used

- React.js
- TypeScript
- HTML5
- CSS3
- Tailwind CSS
- Vite

Responsibilities

- User Registration and Login
- Temple Browsing
- Darshan Ticket Booking
- Booking History
- Admin Dashboard
- Responsive User Interface

2. Business Logic Layer

The business logic layer processes user requests, validates data, performs authentication, and communicates with the database.

Technologies Used

- Node.js
- Express.js
- JWT Authentication
- bcrypt

- RESTful APIs

Responsibilities

- User Authentication
- Authorization
- Temple Management
- Booking Management
- Slot Management
- Admin Operations
- API Validation

3. Database Layer

The database layer stores all application information securely.

Technology Used

- MongoDB
- Mongoose ODM

Collections

- Users
- Temples
- Darshan Slots
- Bookings
- Poojas

Data Flow

1. User accesses the React application.
2. User registers or logs into the system.
3. JWT token is generated after successful authentication.
4. User browses available temples.
5. User selects a temple and darshan slot.

6. Booking request is sent to the Express.js server.
7. Backend validates the request.
8. Booking details are stored in MongoDB.
9. Booking confirmation is returned to the frontend.
10. Administrator manages temples, bookings, users, and slots through the admin dashboard.

Key Architectural Features

- Three-Tier Architecture (Presentation, Business Logic, Database)
- MERN Stack Implementation
- RESTful API Communication
- JWT-Based Authentication
- Password Encryption using bcrypt
- MongoDB NoSQL Database
- Responsive User Interface
- Modular Folder Structure
- Scalable Design for Future Enhancements
- Cloud Deployment Ready

Benefits of the Solution Architecture

- **Scalability:** Supports future features such as payment gateways, QR code verification, and AI-based crowd prediction.
- **Security:** Implements JWT authentication, password hashing, and protected APIs.
- **Maintainability:** Modular architecture simplifies development and maintenance.
- **Performance:** Efficient API communication and optimized MongoDB queries improve response times.
- **Reliability:** Separate layers ensure stability and reduce dependencies between components.
- **Cloud Readiness:** Can be deployed using Vercel, Render, and MongoDB Atlas for high availability.

Conclusion

The proposed solution architecture for **DarshanEase – Temple Darshan Ticket Booking System** follows industry-standard MERN Stack design principles. The separation of the presentation layer, business logic layer, and database layer ensures security, scalability, maintainability, and high performance. This architecture provides a strong foundation for delivering a reliable online temple darshan booking platform while supporting future enhancements and cloud deployment.